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Deployment & Configuration

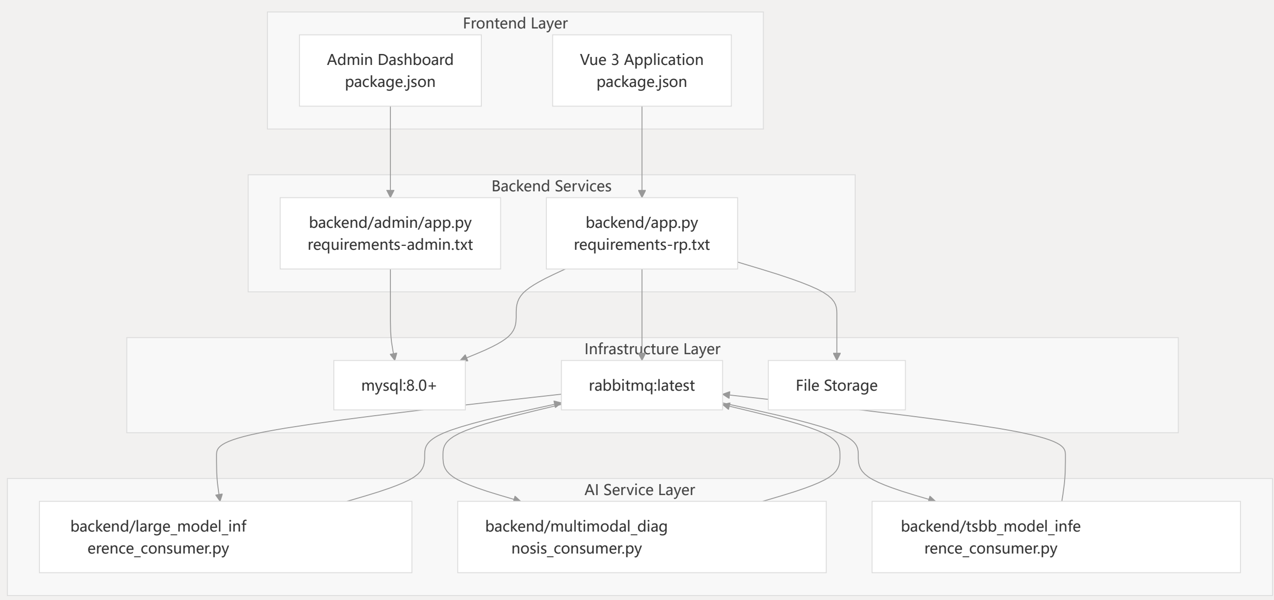
> Relevant source files

This document covers the system dependencies, configuration requirements, deployment architecture, and licensing compliance for the MKTY Smart Healthcare System. It provides the technical foundation needed to deploy and operate the system's microservices architecture.

For information about individual AI services implementation, see AI & Machine Learning Services. For database schema and data management details, see Database & Data Management.

System Architecture Overview

The MKTY system employs a distributed microservices architecture requiring coordination between multiple Python services, AI model inference engines, message queues, and database systems.



Deployment Architecture Overview

Sources: `backend/requirements-rp.txt` 1-17 `backend/requirements-admin.txt` 1-4 **system architecture diagrams**

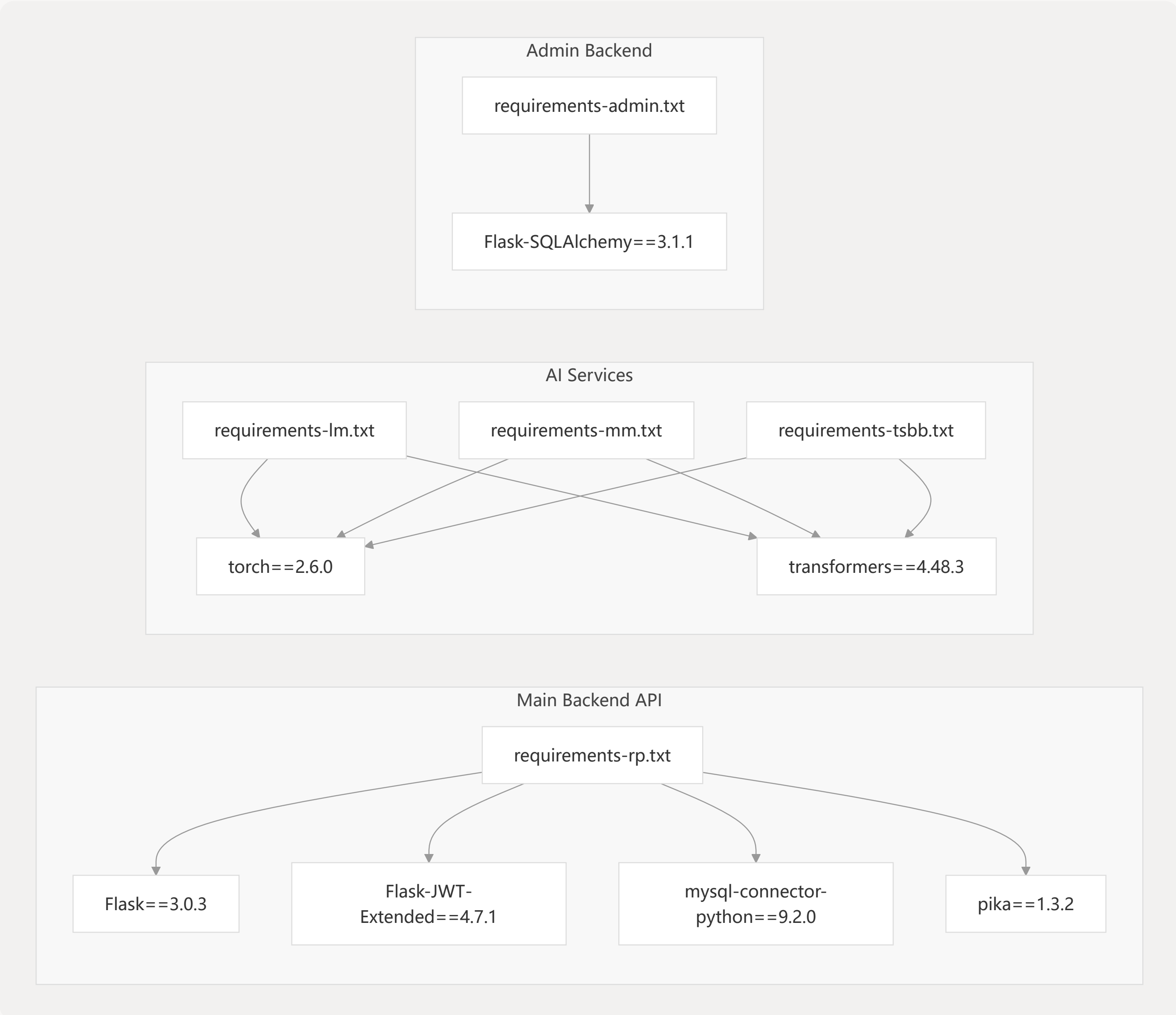
System Dependencies & Requirements

Core Infrastructure Requirements

Component	Version	Purpose
Python	3.8+	Runtime for all backend services
MySQL	8.0+	Primary database with JSON support
RabbitMQ	Latest	Message queue for AI service communication
Node.js	16+	Frontend build system
CUDA	11.8+	GPU acceleration for AI models

Python Dependencies by Service

The system uses service-specific requirement files to manage dependencies for each microservice:



Service-Specific Dependencies Mapping

Sources: `backend/requirements-rp.txt` 1-17 `backend/requirements-lm.txt` 1-4 `backend/requirements-mm.txt` 1-7 `backend/requirements-tsbb.txt` 1-4 `backend/requirements-admin.txt` 1-4

Main Backend Service (requirements-rp.txt)

The primary API service requires the most comprehensive dependency set:

Package	Version	Purpose
`Flask`	3.0.3	Web framework for REST API
`Flask-Cors`	5.0.0	Cross-origin resource sharing
`Flask-JWT-Extended`	4.7.1	JWT authentication
`mysql-connector-python`	9.2.0	MySQL database connectivity
`pika`	1.3.2	RabbitMQ client
`argon2-cffi`	21.3.0	Password hashing
`rank-bm25`	0.2.2	Text search algorithms

Sources: `backend/requirements-rp.txt` 1-17

AI Model Services

All AI services share common PyTorch dependencies:

Package	Version	Purpose
`torch`	2.6.0	Deep learning framework
`transformers`	4.48.3	Hugging Face model library
`pika`	1.3.2	Message queue communication

Multimodal Service Additional Dependencies:

- `open-clip-torch==2.23.0` - Vision-language models
- `Pillow==10.4.0` - Image processing

Sources: `backend/requirements-lm.txt` 1-4 `backend/requirements-mm.txt` 1-7 `backend/requirements-tsbb.txt` 1-4

Admin Backend Service

Minimal Flask setup for administrative functions:

Package	Version	Purpose
`Flask-SQLAlchemy`	3.1.1	ORM for database operations
`PyJWT`	2.8.0	JWT token handling

Sources: `backend/requirements-admin.txt` 1-4

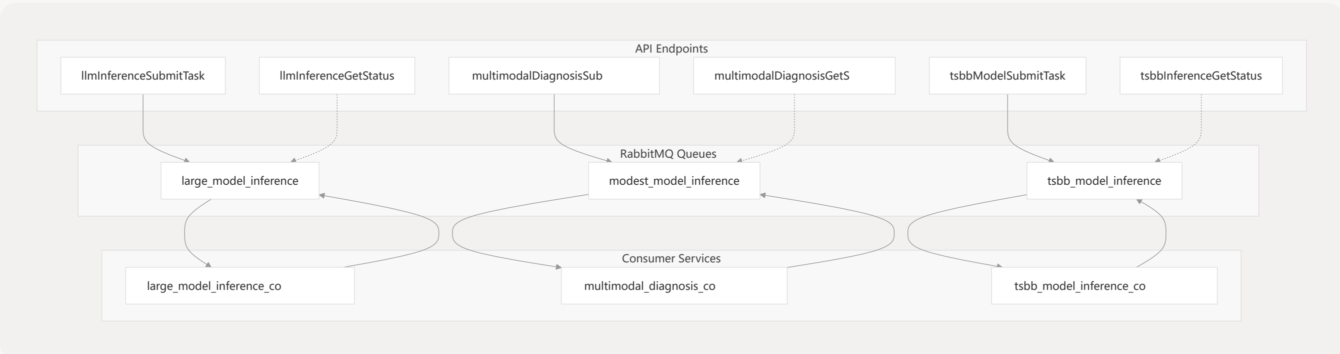
GPU Memory Requirements

Service	Model	VRAM Requirement
`large_model_inference_consumer.py`	MKTY-3B-Chat	8GB
`multimodal_diagnosis_consumer.py`	BioMedCLIP + MarianMT	2GB
`tsbb_model_inference_consumer.py`	BigBird + GRU	2GB

Configuration Architecture

Message Queue Configuration

The system uses RabbitMQ queues for asynchronous AI model communication:



Message Queue Architecture and API Integration

Sources: System architecture diagrams, API endpoint naming patterns from high-level architecture

Database Configuration

MySQL 8.0+ is required for JSON field support used throughout the application:

Configuration	Requirement	Purpose
JSON Support	MySQL 8.0+	User preferences, medical records
Character Set	utf8mb4	Unicode support for multilingual content
InnoDB	Required	Transaction support and foreign keys

Deployment Setup

Service Startup Sequence

1. **Infrastructure Layer**: MySQL, RabbitMQ
2. **AI Model Services**: GPU-dependent consumers first
3. **Backend APIs**: Main and admin services
4. **Frontend**: Vue.js applications

Environment Variables

Variable	Service	Purpose
`MYSQL_HOST`	All backend services	Database connection
`RABBITMQ_URL`	AI services + main API	Message queue connection
`JWT_SECRET_KEY`	Authentication services	Token signing
`CUDA_VISIBLE_DEVICES`	AI services	GPU allocation

Port Configuration

Service	Default Port	Configuration File
Main API	5000	`backend/app.py`
Admin API	5001	`backend/admin/app.py`
Frontend	3000	`package.json`
MySQL	3306	System default
RabbitMQ	5672	System default

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Requirement	Details
Reciprocal Licensing	Modified files must remain open source under MPL-2.0
Attribution Required	Must acknowledge project name, repository link, and author
Transparency	Cannot obfuscate or conceal open source nature

Sources: `LICENSE 1-423`

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